

- $$12 \{1\} \rightarrow \{1\}$$

$$\text{ans} = 1 \times 1 \times 2^4 \times 1$$

$$= 2 \text{ Ans.}$$

Q. palindromic $\rightarrow 1^2 1 \leftarrow$

$$n = 121$$

$$x_e v = 12 L$$

$$\eta = 123$$

$$8r v = 321$$

$$(\eta = \gamma p v)$$

$$(n! = \text{rev})$$

↳ Print (true)

→ Print (false);

int n = i/p ();

```
int num = n;
```

→ int rev = 0;

while ($n > 0$) {

→ int rem = num % 10;

$$r_{EV} = r_{EV} \times 10 + r_{EM};$$
$$\rightarrow num = num / 10;$$

```
if (rev == an)
    cout << "True";
```

```
else {
    sent ("false");
}
```

3

✓ 2



$$\begin{array}{ccc} 2 & 2 & 1 \\ \hline & & \text{int} \end{array} \leftarrow \begin{array}{c} 12 \cdot 3 \\ \hline 12 \end{array}$$

$1\frac{1}{4} \quad 10 \overline{) 12312}$
 $\quad \quad \quad 20$
 $\quad \quad \quad \underline{3}$
 $\quad \quad \quad 2$

$$r_{\rho\nu} = 3^2$$

$$3 \times 10 + 2$$

$$L = 32$$

$$\gamma_{PV} = 3$$

$32^0 + 7$

32 /

$$\begin{aligned} 3 \times 10 &= 30 + 2 \\ &= 32 \end{aligned}$$

$$\begin{aligned} \cdot 1 \quad Y_{2V} &= 32 \times 16 + 1 \\ &= 322 \text{ Ans.} \end{aligned}$$

$$h = \begin{matrix} & 1 & 2 & 1 \\ \begin{matrix} 1 \\ 2 \\ 1 \end{matrix} & \end{matrix}$$

$$\gamma^{\rho\nu} = 121$$

Q2. 

$$0/p \rightarrow 8$$

```
int n = ip[1];
```

```
int num = n;
```

Int ans = 0; $\rightarrow$ // bucket.

... 1 1 0 1 num > 0 }\$ ~~num = 0~~

```

int ans = 0; // bucket.
while (num > 0) {
    int rem = num % 10; // 1/4
    if (rem > ans) {
        ans = rem;
    }
    num = num / 10;
}
return ans;

```

→ 1 2 3 8 5 {4} ans = 0;
 rem = 4

if (rem > ans) ans = rem;
 ans = 4

→ num / 10;

1 2 3 8 5 → ans = 4

1 2 3 8 5 4

→ ans = 5

1 2 3 8
 rem = 8 ans = 5

ans = 8

1 2 3
 rem = 3, ans = 8

1 2
 rem = 2, ans = 8

L

rem = 1, ans = 8

ans = 8

Q. Prime numbers
 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...

" $\rightarrow \{1, \dots, n\}$

$n = 5$ $rem = 1$ $2 \mid 5 \rightarrow 2$

$\rightarrow 5/2$ $\rightarrow 5/3$ $\rightarrow 5/4$

10, 12 $\rightarrow 1$ 2 3 4 5

$\rightarrow 5$ (Prime)

$5/2 \times$

$5/3 \times$

$5/4 \times$

$5/5 \times$

$n = 6$ $\rightarrow$ prime

$\rightarrow$ prime

$n/2 \rightarrow \checkmark$

`int n = ip();`

`Boolean isPrime = true;`

`for (int i = 2; i < n; i++) {`

`if (n/i == 0) isPrime = false;`

`}`

`cout (isPrime);`

double

`Math.sqrt`

2, ..., n-1

`(int) Math.sqrt()`

$n = 6$

$\{2, 3, 4, 5, 6\}$

$2, \dots, n-1$
 $n = 6, 5$
 $\frac{6}{4}$
 $2, 3, 4$
 $\boxed{\text{isPrime} = \text{false}}$
 $\rightarrow (\text{int}) \text{math.sqrt}()$

$2, \dots, n$

$n \rightarrow 10^6$
 $\rightarrow \text{observation}$

$\text{time} \times \text{space} \xrightarrow{\text{complexity}}$

$2 \quad 3 \quad 4 \quad 5 \quad 6$

$\text{num} / 2 \rightarrow$

$6/2 \rightarrow 3$

$\sqrt{n} \quad n = 36$

$\rightarrow 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 9 \quad 10 \quad 11 \quad 12 \quad 13 \quad 14 \quad 15$

$$2 \times 8 = 16$$

$$8 \times 2 = 16$$

$$\boxed{16/2 = 8}$$

4×4

$$\rightarrow 2 \times 18$$

$$\rightarrow 4 \times 9$$

$$\rightarrow 6 \times 6$$

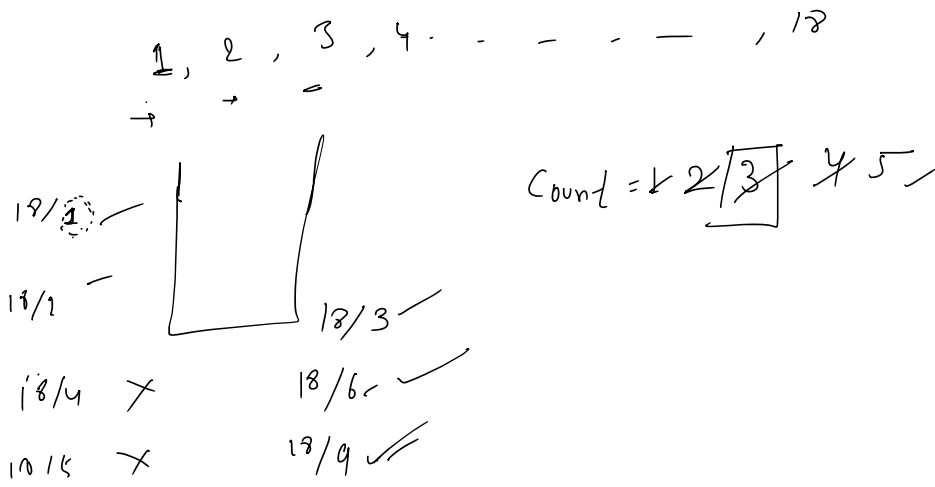
$$\rightarrow 9 \times 4$$

$$\rightarrow 18 \times 2$$

$\downarrow$
 $?$

Q. 18

1, 2, 3, 6, 9, 18



Alternative

1, 2, 3, 4, 5, 6, 7, 8, 9, ..., 18

4x4 = 16
5x4 = 20

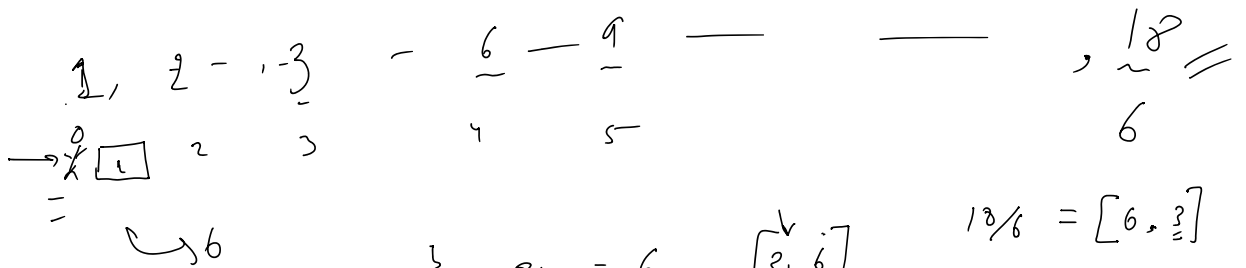
$$18/1 = [1, 18] ✓$$

$$18/2 = 9 [2, 9]$$

$$18/3 = 6 [3, 6] ✓$$

$$\sqrt{18} = 4$$

$$18/6 = 3 [6, 3] ✓$$



[1, 18]

18/1 ✓

1 2 3



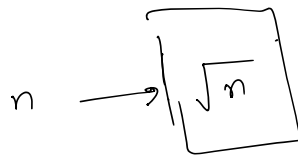
$$18/3 = 6 [3, 6]$$

$$18/6 = [6, 3]$$

$$18/2 = [2, 9]$$

Print(count);

$$\sqrt{16} = 4$$



Q H.C.D = Greatest Common Divisor.

Ans $\rightarrow$ H.C.F

(2, 18)

$$\begin{array}{r} \sqrt{2} \quad 18 \quad | \quad 9 \\ \underline{18} \\ \times \end{array}$$

H.C.F
1, 2, 3, 4, ... 12

2/1, 18/1

$\rightarrow$
2/2, 18/2

$\rightarrow$ 1, 9



2/3

H.C.F $\rightarrow$ 2

$n = \min(p, q)$

$n = 2$

H.C.F

1, 2

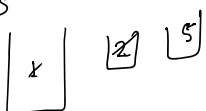
Smaller
H.C.F (n, m)

H.C.F $\rightarrow$ n

H.C.F (25, 35)

1, 2, ..., 25

ans



ans $\rightarrow$ curr_number

ans = curr_number;

int a = i/p;

int b = i/q;

int n = min(a, b);

int ans = 0;

for (int i = 1; i <= n; i++) {

1, 2, 3, 4, 5, 6, 7, ... 25

```

int ans = 0;
for (int i = 1; i <= n; i++) {
    if (a/i == 0 && b/i == 0) {
        ans = i;
    }
}
cout << ans;

```

→ 2 5 , 3 5
 - 5 5
 ✓ ✓

→ i = 5
 → i = 6
 →

ans = 5

ans = 1